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Python Programming
Assignment - 1

1) who declared Python and why was it created? Explain one major difference between Python & C in terms of program execution.

Ans: Python programming Language was created by - Guido van Rossum. The creation was started in year 1989 and the first ~~reed~~ official release was ~~at~~ in the year 1991. It was developed at Centrum Wiskunde & Informatica (CWI), Netherlands.

The major difference between Python & in terms of program execution is Python is an Interpreted Language where as C is a Compiled Language.
Python code is executed line-by-line using an interpreter This makes debugging easy because errors are shown immediatly.

2) Explain the Python build process starting from .py file to execution. Mention the role of:
- ~~B~~ Python Interpreter
- Byte code
- PVM.

Ans: a) Developer ~~ce~~ creates the file with the extension .py on IDE and write/type source code in it and saves it.

b) After that Lexical Analysis is done. This means that the python interpreter scans the code and breaks it into Tokens. ~~which~~

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c) Then, the tokens are parsed and transformed into Abstract Syntax Tree (AST) in which it checks for any syntax errors.

d) Ahead, AST is compiled into Python ~~byted~~ bytecode which creates an internal file with the extension .pyc files. This file is stored in --pycache-- folder and this Bytecode file is platform independent.

e) After that, Python Virtual Machine (PVM) reads this bytecode and execute it line-by-line. It handles runtime actions like memory ~~allocation or~~ management, method calls and control flows.

3) What is bytecode in Python? Where is it ~~sto~~ stored and why does Python use it?

Ans: Bytecode is a low-level language. Python is a high level programming language but the machine or microprocessor do not understand ~~or~~ it. ~~therefore,~~ ~~& the~~ thats why, to make it understand the Python code is compiled into bytecode ~~so~~ using AST so it should be understandable by the PVM because machine understands low-level language. Also bytecode is plat-form independent.

Byte code is stored as .pyc files in the --pycache-- directory. Python use bytecode because it is a low-level language set of instructions which understandable by the PVM so that PVM executes its bytecode line-by-line.

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4) Is Python a complied language or an interpreted language? Justify your answer with execution steps.

Ans:- Python is an interpreted language. Because, ~~inter~~ interpreter execute the Python code line-by-line and this makes debugging easy because ~~ex~~ errors are shown immediatly.

5) ~~whi~~ What is the purpose of the --pycache-- folder?
Ans:- The purpose of the --pycache-- folder is it stores bytecode ~~.pyc~~ .pyc files in it.

6) Python is called a dynamically typed language. Explain this with a short code example.
Ans:- In Python, you don't need to declare the type of variable while creating it because it automatically assigned during runtime based on the value assigned and also we can even change the type of variable by assigning a new value of different type.
~~Let~~ For example:

```
x = 11              # x is an integer
x = "Jay Ganesh..."  # Now x is a string
```

If we take comparitively, in C programming we have to declare datatype in code but in Python we do not have to
For example: below is C program

```
int x = 11;          // x is an integer
```

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7) Explain why Python is considered:
- Platform independent
- Portable

Ans: ~~Platform independent :-~~
Python is considered as Platform independent because Python programs can run on Windows, Linux & macOS without changing the code. Thats why, this makes Python portable across different systems.

8) What do you understand by "Indentation - specific language"? why is indentation mandatory in Python?

Ans: Indentation - specific language ~~means the language in which indentation is used~~ is one where the leading spaces or tabs is ~~u~~ used at the beginning of the lines which dictate or ~~th~~ to structure the code under any function.
Indentation mandatory in Python because it is its syntatic requirement used to define the structure and scope of the code. Unlike other languages like C, C++, java, ~~that~~ these are blocked ~~stur~~ structured and used curly brackets '{}' to define the scope of the program.

9) What will happen if indentation is not maintained properly? Explain with a small code snippet.

Ans: If indentation is not maintained properly in Python, the interpreter will raise an IndentationError. Unlike many other languages we curly braces {} to define code blocks. Python uses whitespaces (indentation) to determine grouping of statements.

For example :

```
x = 10
if x > 5:
print ("x is greater than 5")
```

Output will be:

```
File "D:\Python\IndentationError.py", line 4
  print ("x is greater than 5")
  ^^^^^
IndentationError: expected an idented block after 'if' statement
on line 3
```

The correct way to write this code is -

```
x = 10
if x > 5:
    print ("x is greater than 5")
```

Output :

```
x is greater than 5
```

10) Python supports multiple programming paradigms. Name any three paradigms and explain one in brief.

Ans: Python supports :

- Procedural Programming
- Object - oriented Programming
- Functional Programming

This makes python flexible for different types of applications

Object-oriented pradigm is one of the most common ~~a~~ because it uses the concepts of classes, objects and inheritance among others to structure the program.